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Patent Public Search | Text View

United States Patent Application Publication

20250279733

Kind Code

A1

Publication Date

September 04, 2025

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SYSTEMS AND METHODS FOR ACTIVE AND PASSIVE COOLING OF ELECTRICAL COMPONENTS

Abstract

A power module, includes: one or more power switches; a passive heat dissipation element in contact with and facing a first surface of the one or more power switches, the passive heat dissipation element having a first planar surface and a second planar surface; and an active heat dissipation element in contact with and facing a second surface of the one or more power switches, wherein the active heat dissipation element is positioned opposite the passive heat dissipation element relative to the one or more power switches and is in thermal contact with the passive heat dissipation element.

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Appl. No.: 19/211665

Filed: May 19, 2025

Related U.S. Application Data

parent US continuation 18162430 20230131 parent-grant-document US 12334852 child US 19211665

us-provisional-application US 63378601 20221006

us-provisional-application US 63377512 20220928

us-provisional-application US 63377501 20220928

us-provisional-application US 63377486 20220928

Publication Classification

Int. Cl.: H02M7/5387 (20070101); B60L3/00 (20190101); B60L15/00 (20060101); B60L15/08 (20060101); B60L15/20 (20060101); B60L50/40 (20190101); B60L50/51 (20190101); B60L50/60 (20190101); B60L50/64 (20190101); B60L53/20 (20190101); B60L53/22 (20190101); B60L53/62 (20190101); B60R16/02 (20060101); G01R15/20 (20060101); G06F1/08 (20060101); G06F13/40 (20060101); H01L21/48 (20060101); H01L23/00 (20060101); H01L23/15 (20060101); H01L23/367 (20060101); H01L23/373 (20060101); H01L23/40 (20060101); H01L23/467 (20060101); H01L23/473 (20060101); H01L23/495 (20060101); H01L23/538 (20060101); H01L25/00 (20060101); H01L25/07 (20060101); H02J7/00 (20060101); H02M1/00 (20070101); H02M1/08 (20060101); H02M1/084 (20060101); H02M1/088 (20060101); H02M1/12 (20060101); H02M1/32 (20070101); H02M1/42 (20070101); H02M1/44 (20070101); H02M3/335 (20060101); H02M7/00 (20060101); H02M7/537 (20060101); H02M7/5395 (20060101); H02P27/06 (20060101); H02P27/08 (20060101); H02P29/024 (20160101); H02P29/68 (20160101); H03K19/20 (20060101); H05K1/14 (20060101); H05K1/18 (20060101); H05K5/02 (20060101); H05K7/20 (20060101); H10D64/01 (20250101)

U.S. Cl.:

CPC H02M7/5387 (20130101); B60L3/003 (20130101); B60L15/007 (20130101); B60L15/08 (20130101); B60L50/40 (20190201); B60L50/51 (20190201); B60L50/60 (20190201); B60L50/64 (20190201); B60L53/20 (20190201); B60L53/22 (20190201); B60L53/62 (20190201); B60R16/02 (20130101); G01R15/20 (20130101); G06F1/08 (20130101); G06F13/4004 (20130101); H01L21/4882 (20130101); H01L23/15 (20130101); H01L23/3672 (20130101); H01L23/3675 (20130101); H01L23/3735 (20130101); H01L23/4006 (20130101); H01L23/467 (20130101); H01L23/473 (20130101); H01L23/49562 (20130101); H01L23/5383 (20130101); H01L24/32 (20130101); H01L24/33 (20130101); H01L25/072 (20130101); H01L25/50 (20130101); H02J7/0063 (20130101); H02M1/0054 (20210501); H02M1/08 (20130101); H02M1/084 (20130101); H02M1/088 (20130101); H02M1/123 (20210501); H02M1/32 (20130101); H02M1/322 (20210501); H02M1/327 (20210501); H02M1/4258 (20130101); H02M1/44 (20130101); H02M7/003 (20130101); H02M7/537 (20130101); H02M7/53871 (20130101); H02M7/53875 (20130101); H02M7/5395 (20130101); H02P27/06 (20130101); H02P27/08 (20130101); H02P27/085 (20130101); H02P29/024 (20130101); H02P29/027 (20130101); H02P29/68 (20160201); H05K1/145 (20130101); H05K1/181 (20130101); H05K1/182 (20130101); H05K5/0247 (20130101); H05K7/20154 (20130101); H05K7/2039 (20130101); H05K7/2049 (20130101); H05K7/20854 (20130101); H05K7/209 (20130101); H05K7/20927 (20130101); H10D64/018 (20250101); B60L15/20 (20130101); B60L2210/30 (20130101); B60L2210/40 (20130101); B60L2210/42 (20130101); B60L2210/44 (20130101); B60L2240/36 (20130101); G06F2213/40 (20130101); H01L2023/405 (20130101); H01L2023/4087 (20130101); H01L2224/32225 (20130101); H01L2224/32245 (20130101); H01L2224/33181 (20130101); H02J2207/20 (20200101); H02M1/0009 (20210501); H02M3/33523 (20130101); H02P2207/05 (20130101); H03K19/20 (20130101); H05K2201/042 (20130101); H05K2201/10166 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of priority to U.S. patent application Ser. No. 18/162,430, filed Jan. 31, 2023, which claims priority to U.S. Provisional Patent Application No. 63/377,486, filed Sep. 28, 2022, U.S. Provisional Patent Application No. 63/377,501, filed Sep. 28, 2022, U.S. Provisional Patent Application No. 63/377,512, filed Sep. 28, 2022, and U.S. Provisional Patent Application No. 63/378,601, filed Oct. 6, 2022, the entireties of which are incorporated by reference herein.

TECHNICAL FIELD

[0002] Various embodiments of the present disclosure relate generally to power switch cooling, and, more particularly, to combinations of passive and active cooling systems.

BACKGROUND

[0003] Automotive cooling systems often use one or more heat sinks to dissipate heat from power switches. Heat sinks provide effective means of dissipating heat by thermally conducting heat into a fluid medium, often via a Thermal Interface Material (TIM). Generally, a series of individual spring clamps provide the pressure required to compress the TIM over the product's lifetime.

[0004] The background description provided herein is for the purpose of generally presenting the context of the disclosure. Unless otherwise indicated herein, the materials described in this section are not prior art to the claims in this application and are not admitted to be prior art, or suggestions of the prior art, by inclusion in this section.

SUMMARY OF THE DISCLOSURE

[0005] In some aspects, the techniques described herein relate to a power module, including: one or more power switches; a passive heat dissipation element in contact with and facing a first surface of the one or more power switches, the passive heat dissipation element having a first planar surface and a second planar surface; and an active heat dissipation element in contact with and facing a second surface of the one or more power switches, wherein the active heat dissipation element is positioned opposite the passive heat dissipation element relative to the one or more power switches and is in thermal contact with the passive heat dissipation element.

[0006] In some aspects, the techniques described herein relate to a power module, wherein the passive heat dissipation element is shaped to be enclosure-free.

[0007] In some aspects, the techniques described herein relate to a power module, wherein each surface of the passive heat dissipation element terminates in free space.

[0008] In some aspects, the techniques described herein relate to a power module, wherein the one or more power switches are mechanically connected to at least one of the active heat dissipation element or the passive heat dissipation element via a clamp force.

[0009] In some aspects, the techniques described herein relate to a power module, wherein the passive heat dissipation element includes a thermo-conductive plate.

[0010] In some aspects, the techniques described herein relate to a power module, wherein the passive heat dissipation element includes a first end and a second end, and wherein the first end and the second end are in thermal contact with the active heat dissipation element.

[0011] In some aspects, the techniques described herein relate to a power module, wherein passive heat dissipation element includes a third planar surface on a same plane as the first planar surface, wherein the second planar surface is on a different plane as the first planar surface and connects the first planar surface to the third planar surface.

[0012] In some aspects, the techniques described herein relate to a power module, further including a first curved surface connecting the first planar surface to the second planar surface and a second curved surface connecting the third planar surface to the second planar surface.

[0013] In some aspects, the techniques described herein relate to a power module, wherein the second planar surface is in direct contact with the active heat dissipation element.

[0014] In some aspects, the techniques described herein relate to a power module, wherein the second planar surface is between a first power switch of the one or more power switches and a

second power switch of the one or more power switches.

[0015] In some aspects, the techniques described herein relate to a power module, further including an inlet and an outlet wherein the active heat dissipation element includes an enclosure shaped for fluid passage, the enclosure further shaped to receive fluid from the inlet and to expel fluid via the outlet.

[0016] In some aspects, the techniques described herein relate to a power module, wherein the power module is configured for fluid flow only via the active heat dissipation element.

[0017] In some aspects, the techniques described herein relate to a method for dissipating heat, the method including: generating heat at a power switch; dissipating at least a portion of the heat generated at the power switch to a passive heat dissipation element in contact facing a first surface of the power switch, wherein each surface of the passive heat dissipation element terminates in free space; and dissipating at least a portion of the heat dissipated to the passive heat dissipation element to an active heat dissipation element in contact with and facing a second surface of the power switch via thermal contact between the passive heat dissipation element and the active heat dissipation element, wherein the active heat dissipation element includes a fluid passage.

[0018] In some aspects, the techniques described herein relate to a method, wherein the passive heat dissipation element is shaped to be enclosure-free.

[0019] In some aspects, the techniques described herein relate to a method, further including dissipating at least a portion of the heat dissipated to the active heat dissipation element to a fluid in the fluid passage.

[0020] In some aspects, the techniques described herein relate to a method, wherein the at least the portion of the heat dissipated from the passive heat dissipation element to the active heat dissipation element is dissipated via one or more contact points thermally connecting the passive heat dissipation element and the active heat dissipation element.

[0021] In some aspects, the techniques described herein relate to a power module, including: a heat generating electrical component; and a passive heat dissipation element in contact with and facing a first surface of the heat generating electrical component, the passive heat dissipation element including: a clamp structure; a contact interface; a planar surface facing the heat generating electrical component; a first curved surface connecting the planar surface to the clamp structure; and a second curved surface connecting the planar surface to the contact interface, wherein at least one of the clamp structure or the contact interface is in direct contact with an active heat dissipation element.

[0022] In some aspects, the techniques described herein relate to a power module, wherein the passive heat dissipation element is shaped to be enclosure-free.

[0023] In some aspects, the techniques described herein relate to a power module, wherein the active heat dissipation element is an active heat sink having a fluid passage.

[0024] In some aspects, the techniques described herein relate to a power module, wherein each surface of the passive heat dissipation element terminates in free space.

[0025] Additional objects and advantages of the disclosed embodiments will be set forth in part in the description that follows, and in part will be apparent from the description, or may be learned by practice of the disclosed embodiments. The objects and advantages of the disclosed embodiments will be realized and attained by means of the elements and combinations particularly pointed out in the appended claims.

[0026] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the disclosed embodiments, as claimed.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0027] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various exemplary techniques and together with the description, serve to explain the principles of the disclosed techniques.

[0028] FIG. 1 depicts an exemplary plate combined cooling system, according to one or more embodiments.

[0029] FIG. 2 depicts a cross-sectional view of the exemplary plate combined cooling system of FIG. 1, according to one or more embodiments.

[0030] FIG. 3 depicts a side view of the exemplary plate combined cooling system of FIG. 1, according to one or more embodiments.

[0031] FIG. 4 depicts a top view of the exemplary plate combined cooling system of FIG. 1, according to one or more embodiments.

[0032] FIG. 5 depicts a cross-sectional view of an exemplary fin combined cooling system, according to one or more embodiments.

[0033] FIG. 6 depicts a base view of an exemplary fin combined cooling system, according to one or more embodiments.

[0034] FIG. 7 depicts a cross-sectional view of an exemplary fin combined cooling system, according to one or more embodiments.

[0035] FIGS. 8A and 8B depict exemplary plate-fin combined cooling systems, according to one or more embodiments.

[0036] FIG. 9 depicts an internal view of an exemplary fin combined cooling system, according to one or more embodiments.

[0037] FIG. 10 depicts a flowchart for dissipating heat using a passive heat dissipation element and an active heat dissipation element, according to one or more embodiments.

[0038] FIG. 11 depicts a schematic diagram for connecting a cooling structure to an electrical component, according to one or more embodiments.

[0039] FIG. 12 depicts a flowchart for connecting a cooling structure to an electrical component, according to one or more embodiments.

DETAILED DESCRIPTION

[0040] The terminology used herein may be interpreted in its broadest reasonable manner, even though it is being used in conjunction with a detailed description of certain specific examples of the present disclosure. Indeed, certain terms may even be emphasized herein; however, any terminology intended to be interpreted in any restricted manner will be overtly and specifically defined as such in this Detailed Description section. Both the general description and the detailed description are exemplary and explanatory only and are not restrictive of the features, as claimed.

[0041] As used herein, the terms “comprises,” “comprising,” “having,” “including,” or other variations thereof, are intended to cover a non-exclusive inclusion such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent to such a process, method, article, or apparatus. In this disclosure, relative terms, such as, for example, “about,” “substantially,” “generally,” and “approximately” are used to indicate a possible variation of +10% in a stated value. The term “exemplary” is used in the sense of “example” rather than “ideal.” As used herein, the singular forms “a,” “an,” and “the” include plural reference unless the context dictates otherwise.

[0042] An electronic device cooling system may include two active heat sinks on either side (e.g., opposing top and bottom sides) of one or more electrical components, such as power switches. An active heat sink may be a heat sink having components (e.g., fins or compartments) through which a coolant material flows. For simplicity, this disclosure references electrical components as power switches, though it will be understood that the techniques disclosed herein may apply to any

electrical component (e.g., power switches, chips, chip components, controllers, etc.) that may generate heat. Two active heat sinks may provide thermal dissipation by drawing heat away from the one or more power switches and towards a surface area of a power switch, and further towards an active heat sink that makes direct or indirect contact with a cooling medium, e.g., coolant or air, associated with the active heat sinks. As discussed herein, benefits of using a passive cooling structure include reducing the risk of a coolant leak (e.g., onto a power switch), reduced or controlled thermal resistance, reduced complexity, and the like. Accordingly, it may be desirable to improve cooling system function, e.g., by using combinations of active cooling (e.g., via an active heat sink) and passive cooling, alternative active cooling structures, or the like.

[0043] According to an example implementation, a cooling system may include a combination of active and passive cooling systems. For example, a cooling system may include an active cooling system, e.g., an active heat sink, and a passive cooling system, e.g., a passive heat sink having a thermo-conductive cooling structure. The thermo-conductive cooling structure may include a plate of thermally conductive materials, as described in more detail below. The passive heat sink (e.g., thermo-conductive cooling structure) may contact both the active heat sink and a power switch. The passive heat sink (e.g., thermo-conductive cooling structure) may be configured to dissipate heat away from the power switch and toward the active heat sink.

[0044] According to another example implementation, the cooling system may include a fin structure. The fin structure may include fins sintered to one side of each individual power switch, and may be sealed against a power switch housing. This fin structure may be an active cooling system and may be combined with another active cooling system, e.g., an active heat sink, or a passive heat sink, e.g., a thermo-conductive cooling structure. In some implementations, the fin structure may be in direct physical and/or thermal contact with a cooling structure, e.g., an active heat sink or a thermo-conductive cooling structure, and to each power switch substrate of a power module.

[0045] FIG. 1 depicts an exemplary combined cooling system **100**, according to one or more embodiments. The combined cooling system **100** may include one or more power switches **105** (generally referred to as power switch **105**), an active heat sink **110**, which may be a single active heat sink or a combination of adjacent and/or attached heat sinks, and a passive heat sink **115**, which may be a single component or a combination of adjacent and/or attached components. Active heat sink **110** may be an active cooling solution and passive heat sink **115** may be a passive cooling solution. For example, active heat sink **110** may actively dissipate heat to a coolant medium whereas passive heat sink **115** may dissipate some or all heat to active heat sink **110** or another heat dissipation component. As further discussed herein, heat dissipated from power switch **105** may be dissipated via both active heat sink **110** and passive heat sink **115**. Passive heat sink **115** may passively transfer heat to active heat sink **110** via one or more contacts with active heat sink **110**, as further discussed below. According to embodiments of the disclosed subject matter, both active and passive cooling may be implemented by using the passive heat sink **115** and active heat sink **110** disclosed herein.

[0046] Power switch **105** may include, e.g., approximately a 400-Volt or 800-Volt Silicon Carbide Inverter for electrified vehicles, although other suitable structures and components are also contemplated. Power switch **105** may include one or more silicon carbide (SiC)-based power switches that deliver relatively high power densities and efficiencies needed to, for example, extend battery range and performance. Power switch **105** may include circuitry and components that are configured to convert direct current (DC) from the electric vehicle battery to alternating current (AC) current, which can be utilized, for example, within an electric motor that drives a propulsion system. Power switch **105** may be installed on one or more power board assemblies. A power board assembly may include one or more power switches **105** (e.g., six power switches **105**), one or more active heat sinks **110**, one or more passive cooling structures or cooling jackets, and/or mechanical components to ensure the mechanical integrity of the power module assembly. A power board

assembly may include an array of electronic packages and input/output (I/O) devices disposed on a circuit board. The one or more power switches **105** may be of any suitable form, e.g., an electronic power switch, with any suitable configuration, e.g., single pole single throw.

[0047] Power switch **105** may be disposed in or between one or more surfaces that may form a power switch housing. For example, the one or more power switches **105** may be disposed within a housing having a first surface including or facing active heat sink **110** and a second surface including or facing passive heat sink **115**. Power switch **105** may thermally and/or mechanically contact a surface of active heat sink **110** and/or a surface of passive heat sink **115** via any suitable means. These connections may allow for various assembly conditions. For example, power switch **105** may be adhered to the one or more surfaces using soldering or sintering, or may be attached using any mechanical or adhesive component such as a clamp, a spring pressure component, or the like. Power switch **105** may mechanically contact active heat sink **110**, passive heat sink **115**, or both active heat sink **110** and passive heat sink **115**. This contact may enable heat generated from the operation of power switch **105** to dissipate across active heat sink **110** and/or passive heat sink **115**, as described in more detail below.

[0048] A passive cooling system (e.g., passive heat sink **115**) may be different than an active cooling system (e.g., active heat sink **110**) such that the passive cooling system does not directly dissipate heat via a coolant (e.g., a fluid or air). Accordingly, no portion of a passive cooling structure is in direct contact with a coolant in some embodiments. Rather, a passive cooling system may include, for example, a uniform conductive surface that covers (e.g., partially or fully covers) or faces at least one surface of power switch **105**. The uniform conductive surface may be substantially parallel to the at least one surface of power switch **105**. Each power switch **105** may be associated with a corresponding uniform conductive surface of a passive cooling component, as shown in FIG. 1. The surface of power switch **105** may be in contact with the uniform conductive surface of the passive cooling system. Further, given that a passive cooling system (e.g., a passive heat sink) may not directly contact a coolant, the shape of a cooling system including one or more thermo-conductive cooling systems may be customizable. For example, an active cooling system may be designed to have a coolant medium through portions of the active cooling structure. A passive cooling structure may allow design flexibility as its design may not be limited by coolant flow considerations. Accordingly, a passive cooling system may provide greater flexibility of components, dimensions, and/or locations in comparison to active cooling systems.

[0049] FIG. 2 depicts a cross-sectional view of an exemplary combined active and passive cooling system **100**. As shown in greater detail in FIG. 2, active heat sink **110** may provide a thermal interface for power switch **105** such that heat radiated from power switch **105** is dissipated via active heat sink **110** (e.g., directly or via passive heat sink **115**, as further discussed herein). Active heat sink **110** may include multiple components. The material of active heat sink **110** may be selected based on an applicable thermal performance to cool the power switches **105**. For example, active heat sink **110** may include copper, aluminum alloys, other conductive materials, etc. or combinations thereof. In some embodiments, active heat sink **110** may be similar in operation to a radiator used in an internal combustion engine. Active heat sink **110** may be configured to have one or more coolant channels **110A**. The one or more coolant channels **110A** may be used to disperse coolant (e.g., fluid or air) to active heat sink **110**. Further, a power module seal may be provided between the one or more coolant channels **110A** and power switch **105** to prevent leakage. Active heat sink **110** may include an inlet (not shown in FIG. 2) and outlet (not shown in FIG. 2) on opposing sides of the active heat sink **110**. The inlet and outlet each may extend through a base plate. The inlet and outlet may be configured to facilitate circulation of coolant via of the one or more coolant channels **110A**. As shown in FIG. 2, active heat sink **110** may be mounted on an opposite side of power switch **105**, relative to passive heat sink **115** (e.g., a top or bottom side).

[0050] Accordingly, active heat sink **110** may dissipate heat generated at power switch **105** (e.g., via coolant channel **110A**). Passive heat sink **115** may be mounted on the opposite side of power

switch **105** as active heat sink **110** and may be in direct physical and thermal contact with active heat sink **110**. Passive heat sink **115** may include any material suitable for conducting and/or dissipating heat, including, but not limited to, copper, aluminum, etc., or a combination thereof. At least a portion of passive heat sink **115** may be in direct contact with active heat sink **110** such that there is no intervening structure between passive heat sink **115** and active heat sink **110**.

Accordingly, heat from power switch **105** may be dissipated via passive heat sink **115** and may further be transferred to active heat sink **110** for further dissipation. Including only one active heat sink facing one surface of power switch **105** (e.g., such that an opposite surface of power switch **105** faces a passive cooling system) may reduce the risk of leaks, e.g., coolant leaks, and may decrease the cost associated with production and/or maintenance of a device, when compared to having two active heat sinks on opposite sides of power switch **105**.

[0051] Active heat sink **110** may include one or more fins **130** that extend into a coolant medium, e.g., air, a refrigerant, water, oil, etc. The coolant channel **110A** may be formed using the one or more fins **130**. The one or more fins **130** may be configured to have a high surface area to maximize heat dissipation into the coolant medium. The one or more fins **130** may have any suitable configuration, including, but not limited to, fins (e.g., cylindrical), straight fins, flared fins, etc. The one or more fins **130** may be composed of any suitable materials or combination of materials, such as copper, aluminum alloys, or a combination thereof. The coolant medium may flow through active heat sink **110** and around the one or more fins **130**. According to an embodiment, the one or more fins may extend perpendicularly away from a surface of power switch **105** that faces active heat sink **110**.

[0052] As shown in FIG. 2, the one or more fins **130** may extend outwardly from power switch **105**, terminating or positioned within a heat sink fin compartment. In one embodiment, power switch **105** may be located on or adjacent to a surface of heat sink fin compartment. The weight of power switch **105** may be structurally supported, at least in part, by the heat sink fin compartment or the active heat sink **110**. A modular approach may be taken by forming a heat sink fin compartment, which may be sealed or welded to a main heat sink compartment.

[0053] The heat sink fin compartment may include or be connected to seals at points where the heat sink fin compartment connects to a housing or other component of active heat sink **110**. The seals may include any suitable material (e.g., high-polymer vinyl methyl polysiloxanes (silicone), rubber, neoprene, nitrile, ethylene-propylene diene monomer (EPDM), fluorocarbons, etc.) or configuration (e.g., O-ring gasket, manifold seal, labyrinth seal, a plate or sheet, an elastomeric seal, etc.). In some embodiments, the one or more fins **130** may extend away from power switch **105**, or the like. Such configurations may create a seal around the fins **130** to prevent the coolant medium of heat sink fin compartment from leaking or escaping.

[0054] FIG. 3 depicts a side view of the exemplary combined cooling system **100** of FIG. 1, and FIG. 4 depicts a top view of the exemplary combined cooling system **100** of FIG. 1. As depicted in FIG. 3 and FIG. 4, passive heat sink **115** may be configured to mechanically contact the one or more power switches **105** and active heat sink **110**. For example, as shown in FIG. 3, a surface **115D** may contact power switches **105**, and one or more contact surfaces such as clamp structure **120** and/or contact interfaces **125** may contact active heat sink **110** and/or a conductive component in direct contact with active heat sink **110**. According to an embodiment Thermal Interface Material (TIM) may be positioned between one or more contact surfaces such as clamp structure **120** and/or contact interfaces **125** may active heat sink **110** and/or a conductive component in direct contact with active heat sink **110**. Heat may dissipate from a surface of power switch **105** facing the surface **115D** to the passive heat sink **115**. The heat from passive heat sink **115** may dissipate from passive heat sink **115** to active heat sink **110** via clamp structure **120** and/or contact interfaces **125**. Passive heat sink **115** may include one or more components, e.g., a single or multiple pieces of copper or other conductive material that, when joined or connected together, form a contiguous or continuous passive heat sink **115**. For example, a single component passive heat sink **115** may extend across

multiple power switches **105**. Alternatively, for example, multiple segments of passive heat sink **115** may extend across respective power switches **105** and may be joined or connected to each other segment.

[0055] FIG. **3** shows at least one cross-section of power switch **105** surrounded by passive heat sink **115** and active heat sink **110**. Accordingly, heat dissipated by power switch **105** may be dissipated by passive heat sink **115** and/or active heat sink **110**.

[0056] As shown in FIG. **1**, FIG. **2**, and FIG. **3**, passive heat sink **115** may include planar portions **115A** on a first plane that may be connected to respective clamp structures **120** and/or contact interfaces **125** via curved portions **115B**. Different planar portions **115A** may be on a same plane. Planar portions **115A** may on a first plane and clamp structures **120** and/or contact interfaces **125** may be on a second plane. The first plane may be substantially over or above the plane of a surface of power switch **105** that faces passive heat sink **115**. The second plane may be substantially below or parallel to the plane of a surface of power switch **105** that faces away from passive heat sink **115**.

[0057] As shown in FIG. **3**, active heat sink **110** may cover one surface of power switch **105**. Passive heat sink **115** may cover other surfaces of power switch **105**. As shown, surface **115D** of planar portion **115A** of passive heat sink **115** may cover at least a portion of the surface of power switch **105** facing passive heat sink **115**. Curved portions **115B** joining planar portions **115A** to and/or including clamp structure **120** and/or contact interfaces **125** may cover one or more surfaces of power switch **105**.

[0058] TIM may be provided at points where heat is thermally dissipated between one or more component of a power module. For example, as shown in FIG. **3**, TIM **111A** may be provided between clamping structures **120** and active heat sink **110**, TIM **111B** may be provided between passive heat sink **115** and power switch **105**, TIM **111C** may be provided between power switch **105** and active heat sink **110**, and TIM **111D** may be provided between contact interfaces **125** and active heat sink **110**. As also shown in FIG. **3**, passive heat sink **115** may be connected to active heat sink **110** at sinter/solder point **112A** (e.g., to connect clamp structure **120** to active heat sink **110**) and/or sinter/solder point **112B** (e.g., to connect contact interfaces **125** to active heat sink **110**).

[0059] FIG. **4** shows a top view of passive heat sink **115** with at least a portion of the one or more power switches **105** extending outward past passive heat sink **115**. For example, one or more connective features **116** of the one or more power switches may extend outward and do not directly contact passive heat sink **115** and/or active heat sink **110**. The exterior of passive heat sink **115**, as depicted in FIG. **4**, may terminate at either end of the cooling system and connect to active heat sink **110** via one or more clamping structures **120**. Passive heat sink **115** may curve or bend to surround power switch **105** (e.g., on three sides) and to connect passive heat sink **115** to active heat sink **110** via one or more contact interfaces **125**. The curves or bends in passive heat sink **115** may increase the amount of surface area in contact between passive heat sink **115** and active heat sink **110**, e.g., via the one or more contact interfaces **125**, and/or between passive heat sink **115** and power switch **105**. The one or more contact interfaces **125** may form a planar surface along the active heat sink **110**. According to an embodiment, passive heat sink **115** may include an amount of contact interfaces **125** proportionate to the overall size of cooling system **100**, the heat exuded by power switch **105**, etc. For example, a surface area of the one or more contact interfaces may be proportional to the average heat dissipated by the power switch.

[0060] Passive heat sink **115** may be in a bent or curved configuration such that portions of passive heat sink **115** contact active heat sink **110** at either/both ends of a power switch **105**, e.g., via clamp structures **120** and/or contact interfaces **125**. Portions of passive heat sink **115** may mechanically and/or thermally contact power switch **105**. Portions of passive heat sink **115** may mechanically and/or thermally contact active heat sink **110**. The use of clamp structures **120** and/or contact interfaces **125** may provide multiple thermal paths to an active cooling system (e.g., via active heat sink **110**) and/or coolant. In some embodiments, passive heat sink **115** may be mechanically directly connected to active heat sink **110** using clamp structures **120**, e.g., via spring clamps,

sintering, soldering, or other connection mechanisms. According to an embodiment, connecting active heat sink **110** and passive heat sink **115** via a spring loading system may distribute the force across passive heat sink **115**, better enforcing the connection between the structures and applying a clamp force to power switch **105**. Passive heat sink **115** may mechanically and/or thermally contact power switch **105** via a planar surface of power switch **105** (e.g., via TIM) and may mechanically and/or thermally contact active heat sink **110** via one or more clamp structures **120** and/or contact interfaces **125**. The one or more clamp structures **120** and/or contact interfaces **125** may be configured as a planar surface that may extend along a planar surface of active heat sink **110**. The one or more contact interfaces **125** may attach or connect to active heat sink **110** via any suitable method, e.g., laser welding, clamp force, spring loading, etc. The strength of the attachment or connection between the one or more interfaces **125** and active heat sink **110** and/or between the passive heat sink **115** and the one or more power switches **105** may exceed the strength or connection of a brazed copper fluid carrying structure. Accordingly, the passive heat sink **115** may result in a rigid structure to facilitate clamping and distribution of force across multiple power switches **105**.

[0061] A passive cooling structure, such as passive heat sink **115** may be shaped to be enclosure-free. As discussed herein, enclosure-free may refer to a structure that does not include a chamber or a fluid passage. For example, active heat sink **110**, as disclosed herein, include a fluid passage chamber for fluid flow. A passive cooling structure may not include a chamber or other enclosure such that fluid cannot flow through the passive cooling structure. A passive cooling structure may be shaped such each surface of the passive heat dissipation element terminates in free space. For example, as shown in FIGS. **1-4**, each surface of as passive heat sink **115** may be non-overlapping such that no chamber or passage is formed by connecting a portion of as passive heat sink **115** to another portion of as passive heat sink **115**. It will be understood that although each surface of the passive heat dissipation element terminates in free space, one or more portions of passive heat sink **115** may be in contact with or may otherwise connect to other components (e.g., active heat sink **110**). For example, as shown in FIG. **3**, passive heat sink **115** may be in contact with active heat sink **110** at clamp structure **120** and/or contact interfaces **125**. Accordingly, each surface of the passive heat dissipation element may terminate in free space, and one or more portions of the passive heat dissipation element terminating in free space may be in direct and/or thermal contact with one or more other components (e.g., active heat sink **110**).

[0062] There may exist a relationship (e.g., described by Fournier's law or any other applicable transform, algorithm, etc.) between the heat transferred from the power switch **105** to the passive heat sink **115**. In other words, passive heat sink **115** may be configured to dissipate heat from power switch **105** in a linear or non-linear manner. According to an implementation, the configuration of active heat sink **110** and passive heat sink **115** may be sufficient to meet heat dissipation requirements for power switch **105** operating below a threshold current. For example, the threshold current may be approximately 500 amps or lower, approximately 450 amps, approximately 400 amps, or the like.

[0063] While passive heat sink **115** is shown with active heat sink **110** underneath power switch **105** and passive heat sink **115**, it will be understood that the components may be in any suitable configuration. For example, in some embodiments, active heat sink **110** may be configured to sit on the opposite side of power switch **105** than the configuration shown in FIG. **2** (e.g., such that the active heat sink **110** and passive heat sink **115** are swapped). In another example, the components may be configured to mount (e.g., to an engine or vehicle component) in a vertical or angled orientation.

[0064] By adding a passive heat sink (e.g., a thermally-conductive structure) in place of an active second heat sink through which a coolant is actively flowed, production and/or maintenance costs may be significantly reduced. Further, since coolant passes through a single heat sink, the risk of coolant leaks may be mitigated in comparison to coolant passing through multiple heat sinks.

Passive heat sink **115** may further provide structural support to a device based on its rigid form and may be configured to distribute pressure more evenly than active heat sink **110**. For example, passive heat sinks may be more rigid in comparison to active heat sinks (e.g., based on material properties, configuration, and/or one or more contiguous portions). Passive heat sinks may experience reduced deformation in comparison to heat sinks composed of soft annealed copper. [0065] In some embodiments, in lieu of one or more active heat sinks **110**, the one or more fins may be adhered, e.g., sintered, to a base plate such as a direct bond copper structure, the direct bond copper structure disposed between the fins and power switch **105**. In this way, power switch **105** may be in the direct thermal path of the coolant flow to minimize the thermal resistance of the system. Any suitable number of fins may be adhered to the base plate, e.g., approximately 200 fins per power switch **105**. The number and placement of fins **130** may be optimized to ensure proper flow characteristics and sintering capability.

[0066] According to embodiments of the disclosed subject matter, a fin system may be used for active cooling of electronic components such as power switch **105**. FIG. 5 depicts a cross-sectional view of an exemplary cooling system **500**, which may include one or more power switches **105** (generally referred to as power switch **105** herein), an active heat sink **110**, and a fin system **131**. As shown, one or more power switches **105** may be disposed between one or more surfaces. For example, the one or more power switch **105** may be disposed within a housing. A first surface of power switch **105** may face and/or be attached to active heat sink **110** or a passive cooling structure (not shown in FIG. 5) and a second surface (e.g., an opposite surface) of power switch **105** may face and/or be attached to fin system **131**. The power switch **105** may thermally and/or mechanically contact, e.g., be adhered to, attached to, connected to, etc., a surface of active heat sink **110**, a passive cooling structure (e.g., passive heat sink **115**), and/or to a surface of fin system **131** via any suitable means. For example, the one or more power switches **105** may be adhered to the one or more surfaces using soldering or sintering, as described in more detail below. This contact may allow heat generated from the operation of the one or more power switches **105** to dissipate across active heat sink **110**, the passive cooling structure, and/or fin system **131**, as described in more detail below.

[0067] Unlike the continuous fins of some active heat sinks, the fins of fin system **131** may be discrete such that each fin of fin system **131** may be detached from each other fin of fin system **131**. Accordingly, the fins of fin system **131** may be connected by a base plate such that all surfaces of the fins are exposed except for a single surface of each fin, at the base plate. Additionally, the fins of an active heat sink may substantially extend the length of one surface of the active heat sink. The discrete fins of fin system **131** may not extend the length of a surface of a heat sink but, rather, may extend approximately less than 20%, less than 15%, less than 10%, less than 5%, less than 3% or the like of the length of a surface of the heat sink. Additionally, as discussed herein, the discrete fins of fin system **131** may be directly connected to a base plate attached to a power switch **105**, such as via sintering or soldering.

[0068] Fin system **131** (e.g., as shown in FIG. 5 and FIG. 8B) may be positioned in fin housing **145** which includes a fin housing base **145A** and a casting **145B** shown in FIG. 8A. Fin system **131** may include a plurality of fins (e.g., rods) extending away from power switch **105** towards fin housing base **145A**. Power switch **105** may be sealingly isolated from fin system **131** such that coolant (e.g., fluid passed through a fluid passage created between fin system **131** and fin housing **145**) does not leak to power switch **105**. For example, power switch **105** may be sealingly isolated from fin system **131** via an elastomeric seal (e.g., where the elastomeric seal connects casting **145B** to fin housing base **145A**). Fin housing base **145A** may include a surface that faces the fin system **131**. Fin housing base **145A** and casting **145B** may form a chamber through which a coolant may flow. The casting may connect or be in contact with an exterior surface of power switch **105** (e.g., via base plate **119** or other component). Fin housing **145** may include one or more protrusions, as further discussed in reference to FIG. 8A and FIG. 8B, that friction stir weld the exterior surface of

the power switch **105** to the casting. The friction stir weld may generate sufficient friction to sealingly connect power switch **105** to fin housing **145**, to prevent coolant leakage. Fin housing **145** may include one or more protrusions that epoxy attach the exterior surface of the power switch **105** to the casting. The epoxy attachment may provide sufficient sealing to connect power switch **105** to fin housing **145**, to prevent coolant leakage. Fin housing **145** (e.g., via the casting **145B**) may be sealingly attached to the power switch via friction stir welding the fin housing **145** to an exterior surface of power switch **105** (e.g., with or without protrusions). Fin housing **145** (e.g., via the casting **145B**) may be sealing attached to the power switch via epoxy attaching the fin housing **145** to an exterior surface of power switch **105** (e.g., with or without protrusions). Accordingly, an exterior surface of power switch **105**, casting **145B**, and fin housing base **145A** may form a chamber for a coolant medium to flow through to dissipate heat away from fin system **131**.

[0069] FIG. **6** depicts a base view of an exemplary fin combined cooling system. As depicted in FIG. **6**, active heat sink **110** may include an inlet **150a** and outlet **150b**. Inlet **150a** and outlet **150b** may be on opposing sides of active heat sink **110**. Inlet **150a** and outlet **150b** may extend through a base plate **140** such that fluid is received via inlet **150a** and expelled through outlet **150b** for fluid flow via active heat sink **110**. The one or more fin system **131** may be configured to have one or more coolant channels through the fin system **131**. This embodiment may include a single active heat sink **110**, mounted on the opposite side of power switch **105** relative to fin system **131**. Accordingly, heat may be actively dissipated via both an active heat sink **110** and fin system **131**. Including only one active heat sink may reduce the risk of leaks, e.g., coolant leaks, and may decrease the cost associated with production and/or maintenance.

[0070] Alternatively, a passive cooling component (e.g., passive heat sink **115**) may be positioned facing a surface of power switch **105**, as further discussed in reference to FIG. **8A** and FIG. **8B** below. A passive cooling structure may passively dissipate heat (e.g., without a coolant medium) and/or may transfer heat to fin system **131** via one or more contacts the fin system, as further discussed herein. Accordingly, a passive cooling system (e.g., passive heat sink **115**) may be in contact with fin system **131** via a direct or indirect connection. Heat may be dissipated from the passive cooling system to fin system **131** in a manner similar to that discussed above in reference to dissipating heat from a passive cooling system to active heat sink **110**.

[0071] FIG. **7** depicts a cross-sectional view of an exemplary fin combined cooling system **500**. Fin system **131** may include one or more fins **132** that may extend into a coolant medium, e.g., air, a refrigerant, water, oil, etc., as depicted in FIG. **7**. It will be understood that although fins having a cylindrical configuration are shown, the fins may be of any applicable shape such as, for example, pin shaped, wave shaped, rectangular shaped, or oblong shaped, etc. The one or more fins **132** may be configured to have a surface area to maximize heat dissipation into the coolant medium and/or to facilitate coolant flow. The one or more fins **132** may include a first exterior surface in contact with a base plate **119** (as further described herein) such that each other exterior surface of a respective fin is a floating surface. An exterior surface may be any exposed surface of a fin (e.g., exposed to air or a coolant). A floating surface may not be mechanically or electrically connected to any other component and may be floating in free space (e.g., such that each other surface than the first surface may be configured to extend into and/or dissipate heat via contact with a coolant material). The one or more fins **132** may include any suitable materials or combination of materials, such as copper, aluminum alloys, or any other conductive material, or a combination thereof. The coolant medium may flow through fin system **131** and around the one or more fins **132**.

[0072] One or more fins **132** may be mechanically adhered and/or connected directly to one or more power switches **105**. For example, the fins **132** may be connected to a base plate **119**, the base plate in contact with the one or more power switches **105**. In another example, one or more fins **132** may be soldered, sintered, etc. directly to an individual power switch **105**. In another example, the one or more fins **132** may be adhered and/or connected, e.g., soldered, sintered, etc. to a direct bond

copper structure, the direct bond copper structure being disposed between the fins **132** and power switch **105**. In this way, power switch **105** may be in the direct path of the coolant flow via the fins **132**, to minimize its thermal resistance. Any suitable number of fins **132** may be adhered to a power switch **105**, e.g., approximately 200 fins **132** per power switch **105**. The number and placement of fins **132** may be optimized to ensure proper flow characteristics and sintering capability.

[0073] FIG. **8A** and FIG. **8B** depict exemplary fin cooling systems **800**. One or more fins **132** may extend outwardly from power switch **105** into a fin housing **145**, as depicted in FIG. **8A** and FIG. **8B**. In one embodiment, power switch **105** may be attached to or against a surface of fin housing **145**, and its weight may be structurally supported, at least in part, by fin housing **145**. Fin housing **145** may be modular and may include one or more modular components that may be joined (e.g., cased) together. For example, modular components of fin housing **145** may be joined in a structural carrier such as a die cast, which may be sealed or welded to the main housing.

[0074] FIG. **8A** depicts a perspective view of fin housing **145** and FIG. **8B** depicts a cross-sectional view of fin housing **145**. In some embodiments, fin housing **145** may include a seal, in contact with one or more power switch **105**. For example, the base of fin housing **145** may include the die cast, as described above, to form three walls. Base plate **119** (e.g., as shown in FIG. **9**) may connect to the die cast walls to form the boundaries of fin housing **145**. The power switch **105** may be positioned on the base plate **119**, as depicted in FIG. **8A**. The fins **132** may be connected to the base plate **119**. This structure may provide thermal dissipation away from the power switch **105** while preventing fluid from leaking or escaping fin housing **145**.

[0075] In some embodiments, fin housing **145** may include seals at one or more points where base plate **119** connects to fin housing **145**. In some embodiments one or more seals may surround the base plate **119** connected to power switch **105** to prevent coolant from entering power switch **105**. The seals may include any suitable material (e.g., high-polymer vinyl methyl polysiloxanes (silicone), rubber, neoprene, nitrile, ethylene-propylene diene monomer (EPDM), fluorocarbons, etc.) or configuration (e.g., O-ring gasket, manifold seal, labyrinth seal, etc.). One or more protrusions **142** may extend from fin housing **145** in a direction facing power switch **105** to provide friction stir welding or epoxy attachment to the casting to create a coolant seal.

[0076] FIG. **9** depicts an internal view of an exemplary fin cooling system **800**. In some embodiments, the one or more fins **132** may be sealed within fin housing **145** (not shown in FIG. **9**) and/or against power switch **105** (e.g., via a base plate **119**), as depicted in FIG. **9**. A sealing substrate may be positioned between power switch **105** and base plate **119** and may include of one or more plates or sheets of sealing materials, such as silicone, neoprene, EPDM, etc.

[0077] The use of fin system **131**, as disclosed herein, in lieu of a heat sink (e.g., a single or a dual heat sink configuration), may reduce costs and complexity. In some embodiments, costs and complexity may be further reduced by using a passive cooling structure (e.g. a passive heat sink **115**), with a fin system **131**, as depicted in FIG. **8A** and FIG. **8B**. An active and passive combination cooling system including, e.g., passive heat sink **115** and fin system **131**, may provide an even lower cost per product, as well as a reduced risk of leaks. Additionally, passive heat sink **115** may further provide additional structural stability and/or dispersion of force across a given structure, as discussed herein.

[0078] While cooling systems are generally discussed herein with passive heat sink **115** or active heat sink **110** opposite fin system **131**, it should be understood that the components may be in any suitable configuration. For example, in some embodiments, passive heat sink **115** or active heat sink **110** may be configured to on the opposite side of power switch **105** than discussed above (e.g., such that fin system **131** and passive heat sink **115** or active heat sink **110** are swapped). In another example, the device may be configured to such that the components may be in a vertical or angled orientation.

[0079] FIG. **10** depicts a flowchart for dissipating heat using a passive heat dissipation element and

an active heat dissipation element, as discussed herein. At step **1002**, heat may be generated at an electrical component such as power switch **105**. The heat may be generated based on use of the electrical component (e.g., to convert power, to generate power or signals, to transfer power or signals, etc.). At step **1004**, the at least a portion of the heat generated at the electrical component may be dissipated to a passive heat dissipation element (e.g., passive heat sink **115**). The heat may be dissipated via TIM and/or other thermal connection of the electrical component with the passive heat dissipation element. As disclosed herein, a passive heat dissipation element may not include a fluid passage and/or may not provide other coolant based dissipation. The passive heat dissipation element may be shaped such that each surface of the passive heat dissipation element terminates in free space, as discussed herein.

[0080] At step **1006**, at least a portion of the heat received at the passive heat dissipation element may be further dissipated to an active heat dissipation element (e.g., active heat sink **110**). The heat may be dissipated from the passive heat dissipation element to the active heat dissipation element in accordance with the techniques disclosed herein. At step **1008**, at least a portion of the heat received at the active heat dissipation element may be further dissipated via a fluid. The fluid may be flow through a fluid passage of the active heat dissipation element.

[0081] FIG. **11** depicts a technique for connecting (e.g., sintering and/or soldering) a cooling structure to one or more power switches **105**. In various embodiments, as disclosed herein, one or more power module components or devices may be configured for double-sided active cooling (e.g., using two active heat sinks **110**), active and passive cooling (e.g., using an active heat sink **110** and a passive heat sink **115**), or single-sided cooling (e.g., using only an active heat sink **110**), as disclosed herein. A sinter-to-base and/or sinter-on-top process may be implemented to connect a power switch to one or more cooling structures, as further discussed herein. A full assembly (e.g., a power switch **105** assembly and one or more cooling structures) may be dropped-in to a lower casting (e.g., base plate **140**, as shown in FIG. **7**). The full assembly may be adhered to the lower casting (e.g., base plate **140**). Alternatively, or in addition, the full assembly or a portion thereof may be connected to a casting in any applicable manner such as friction stir welding, epoxy attachment, etc. of the full assembly or a portion thereof to a casting. The full assembly may be connected to the casting with TIM, one or more seals **121**, or any other applicable technique. Such connection (e.g., via sintering or soldering) may provide improved thermal resistance for one or more power switches **105**.

[0082] As shown in FIG. **11**, a base substrate **162** may include a top substrate **162a**, a connection material **162b** (e.g., silicon nitride (SiN)), and a bottom substrate **162c**. Top substrate **162a** and bottom substrate **162c** may be formed of a conductive material such as copper, double bonded copper (DBC), etc. Connection material **162b** may be sandwiched between top substrate **162a** and bottom substrate **162c** and may be thermodynamically stable. FIG. **12** depicts a flowchart including steps for the technique for connecting (e.g., sintering or soldering) one or more cooling structures to one or more power switches **105**, as shown in FIG. **11**. At step **1202**, base plate **119** of fin system **131** may be sintered to base substrate **162**. Base plate **119** may be sintered to base substrate **162** at one or more points such that base plate **119** is sealingly secured to base substrate **162**. Base plate **119** may be sintered to base substrate **162** via a sintering layer **174** (e.g., a sintering film, segmented sintering films, sintering strips, sintering portions, etc.).

[0083] At step **1204** of FIG. **12**, one or more components associated with power switch **105** (e.g., drains, die, lead frames, etc.) may be sintered or welded to top substrate **162a** of base substrate **162** and/or to one or more other components of power switch **105**. The one or more components associated with power switch **105** may be sintered to top substrate **162a** of base substrate **162** via sintering layer **176** (e.g., a sintering film, segmented sintering films, sintering strips, sintering portions, etc.). For example, one or more power switch **105** drains and/or a power switch **105** lower lead frame may be sintered to base substrate **162** at one or more points via sintering layer **176**. According to another example, power switch **105** die may be sintered to power switch **105** lower

lead frame and power switch **105** lower lead frame and/or one or more power switch **105** drains may be sintered onto base substrate **162** via sintering layer **176**.

[0084] As shown in FIG. **11**, a base substrate **161** may include a top substrate **161a**, a connection material **161b** (e.g., SiN), and a bottom substrate **161c**. Top substrate **161a** and bottom substrate **161c** may be formed of a conductive material such as copper, double bonded copper (DBC), etc. Connection material **161b** may be sandwiched between top substrate **161a** and bottom substrate **161c** and may be thermodynamically stable. At step **1206** of FIG. **12**, one or more components associated with power switch **105** (e.g., field transistor sources, lead frames, board, die, electrical components, electrical traces etc.) may be sintered to bottom substrate **161c** of base substrate **161**. The one or more components associated with power switch **105** may be sintered to bottom substrate **161c** of base substrate **161** via sintering layer **172** (e.g., a sintering film, segmented sintering films, sintering strips, sintering portions, etc.). For example, a power switch **105** field transistor source and/or upper lead frame may be sintered to base substrate **161** at one or more points via sintering layer **172**. Power switch **105** components **170** such as internal gate drivers and/or decoupling capacitors may have soldered interconnections.

[0085] At step **1208** of FIG. **12**, an active heat dissipation element (e.g., active heat sink **110**) or passive heat dissipation element (e.g., passive heat sink **115**) may be connected to base substrate **161** via any applicable manner including, but not limited to, via soldering or via a TIM. In accordance with the techniques of FIG. **11** and FIG. **12**, one or more internal soldering of a Silicon Carbide (SiC) die may not be required. Accordingly, the base substrate **161** may be soldered to an active heat dissipation element (e.g., active heat sink **110**) or passive heat dissipation element (e.g., passive heat sink **115**) without the use of TIM.

[0086] According to an embodiment, a robotic system may automatically perform the techniques disclosed in FIG. **11** and FIG. **12**. The robotic system may include one or more arms (e.g., having one or multiple degrees of motion), sensors (e.g., motion sensors, position sensors, optical sensors, force sensors, pressure sensors, electrical sensors, etc.), sintering components, soldering components, pick-and-place components, movement mechanisms (e.g., a gantry, a roller, a motor, a joint, etc.), or the like. The robotic system may perform all or some of the techniques disclosed in FIG. **11** and FIG. **12** in response to and/or using a computer program (e.g., via a processor), one or more signals (e.g., sensor signals), a memory, a processor, etc.

[0087] According to an embodiment, one or more fins (e.g., fins **132**) may be attached to individual power switches **105** via base plate **119**, and the one or more fins or corresponding fin system **131** may be separately tested to aid First Time Quality (FTQ). FTQ may be a metric used to measure how easily a given assembly may be manufactured. According to an embodiment, fin system **131** and base plate **119** may be sealingly attached to power switch **105** and/or base plate **140** or other casting via any applicable manner such as an adhesive, an epoxy attachment, and/or friction stir welding, to create a coolant seal.

[0088] The connected power switch **105** and fin system **131**, connected via the techniques described in FIG. **11** and FIG. **12** may be dropped into a casting (e.g., base plate **140**) as discussed herein. Such a configuration may result in an overall height reduction and/or improved sealing of the cooling channel created via fin system **131**. Over-molding of an assembly may be performed after a switch assembly is completed or as an entire assembly is configured in accordance with the techniques described in FIG. **11** and FIG. **12**.

[0089] Other embodiments of the disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. It is intended that the specification and examples be considered as exemplary only, with a true scope and spirit of the invention being indicated by the following claims.

Claims

- 1.** A power module, comprising: one or more power switches; a passive heat dissipation element in contact with and facing a first surface of the one or more power switches, the passive heat dissipation element having a first planar surface and a second planar surface, wherein the passive heat dissipation element comprises a first end and a second end; and an active heat dissipation element in contact with and facing a second surface of the one or more power switches, wherein the active heat dissipation element is positioned opposite the passive heat dissipation element relative to the one or more power switches and is in thermal contact with the passive heat dissipation element wherein the passive heat dissipation element is in direct contact with the active heat dissipation element between the first end and the second end.
- 2.** The power module of claim 1, wherein the passive heat dissipation element is shaped to be enclosure-free.
- 3.** The power module of claim 1, wherein the one or more power switches are mechanically connected to at least one of the active heat dissipation element or the passive heat dissipation element via a clamp force.
- 4.** The power module of claim 1, wherein the passive heat dissipation element comprises a thermally conductive plate.
- 5.** The power module of claim 1, wherein the first end and the second end are on opposite sides of the one or more power switches.
- 6.** The power module of claim 1, wherein the passive heat dissipation element comprises a third planar surface on a same plane as the first planar surface, wherein the second planar surface is on a different plane as the first planar surface and connects the first planar surface to the third planar surface.
- 7.** The power module of claim 6, further comprising a first curved surface connecting the first planar surface to the second planar surface and a second curved surface connecting the third planar surface to the second planar surface.
- 8.** The power module of claim 6, wherein the second planar surface is in direct contact with the active heat dissipation element.
- 9.** The power module of claim 6, wherein the second planar surface is between a first power switch of the one or more power switches and a second power switch of the one or more power switches.
- 10.** The power module of claim 1, further comprising an inlet and an outlet wherein the active heat dissipation element comprises an enclosure shaped for fluid passage, the enclosure further shaped to receive fluid from the inlet and to expel the fluid via the outlet.
- 11.** The power module of claim 1, wherein the power module is configured for fluid flow only via the active heat dissipation element.
- 12.** A method for dissipating heat, the method comprising: dissipating at least a portion of heat generated at a power switch to a passive heat dissipation element in contact facing a first surface of the power switch; and dissipating at least a portion of the heat dissipated to the passive heat dissipation element to an active heat dissipation element in direct contact with and facing a second surface of the power switch via direct thermal contact between the passive heat dissipation element and the active heat dissipation element.
- 13.** The method of claim 12, wherein the passive heat dissipation element is shaped to be enclosure-free.
- 14.** The method of claim 12, further comprising dissipating at least a portion of the heat dissipated to the active heat dissipation element to a fluid in a fluid passage in the active heat dissipation element.
- 15.** The method of claim 12, wherein the at least the portion of the heat dissipated from the passive heat dissipation element to the active heat dissipation element is dissipated via one or more contact points thermally connecting the passive heat dissipation element and the active heat dissipation element.

- 16.** A power module, comprising: a heat generating electrical component; and a passive heat dissipation element in contact with and facing a first surface of the heat generating electrical component, the passive heat dissipation element comprising: a clamp structure; a contact interface; and a planar surface facing the heat generating electrical component, wherein the planar surface is connected to the clamp structure; wherein the clamp structure and the contact interface are in direct contact with an active heat dissipation element.
- 17.** The power module of claim 16, wherein the passive heat dissipation element is shaped to be enclosure-free.
- 18.** The power module of claim 16, wherein the active heat dissipation element is an active heat sink having a fluid passage.
- 19.** The power module of claim 16, wherein the heat generating electrical component is a power switch.
- 20.** The power module of claim 16, wherein the passive heat dissipation element comprises a thermo-conductive plate.
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